Resolving Three Open Problems in Swirl-String Theory via the Electron Compton Frequency

Canonical Derivation of r_c, Γ_0, and $\rho_{\text{core}}/\rho_f$ without Circular Reference to Electromagnetic Constants

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Abstract

Three open problems were identified in the Swirl-String Theory (SST) Route-2 stability programme: (i) the vortex core radius r_c entered the energy functional as an external constant rather than as a derived eigenvalue; (ii) the circulation quantum Γ_0 was defined self-referentially as $2\pi r_c v_{\circlearrowleft}$; and (iii) a factor of $\sim 10^{21}$ separated the Route-2 energy scale from the observed SST mass scale. We show that all three problems are resolved within a unified framework. Problems (i) and (ii) are resolved simultaneously by adopting the electron Compton angular frequency $\omega_C = m_e c^2/\hbar$ as a primitive in place of r_c. Under the replacement $(\rho_f, v_{\circlearrowleft}, r_c) \to (\rho_f, v_{\circlearrowleft}, \omega_C)$, both r_c and Γ_0 become internal predictions of the theory, free of circularity. Problem (iii) is resolved by showing that the core density follows directly from the maximum vortex force and the Planck-scale constraint:

$$\rho_{\text{core}} = \frac{m_e c^2}{2\pi\, v_{\circlearrowleft}^2\, r_c^3},$$

which is derived without invoking ρ_{core} as input. The condensation factor $\mathcal{C} = \rho_{\text{core}}/\rho_f$ then bridges the Route-2 energy scale to the SST mass scale exactly, closing the 10^{21} gap analytically.

Contents

1 Background and Motivation **2**

2 The Compton Frequency as a Primitive **2**
2.1 Definition and SST interpretation . . . 2
2.2 Derivation of r_c without circular reference . . . 3
2.3 Derivation of Γ_0 without circular reference . . . 3
2.4 Status of closure for problems (i) and (ii) . . . 3

3 Identification of the Energy-Scale Gap (Problem iii) **4**
3.1 Magnitude of the discrepancy . . . 4
3.2 Physical interpretation . . . 4
3.3 Derivation of ρ_{core} from the maximum force constraint . . . 5
3.4 Physical interpretation via the virial theorem . . . 6
3.5 Condensation factor in closed form . . . 6

4 Connection to the Fine-Structure Constant **6**

5 Revised Primitive Set and Canon Update **6**

6 Numerical Verification **7**

7 Summary of Resolved and Open Problems **7**

1 Background and Motivation

In SST [1], the swirl-string is modelled as a closed vortex filament in an incompressible, inviscid continuum. The Route-2 stability analysis postulates that the physical core radius r_c is the stationary point of the energy functional

$$\mathcal{E}_K(a) = \rho_f \Gamma_0^2 L_K \left[A_K \ln \frac{L_K}{a} + a_K \right] + \frac{\pi}{2} \rho_f v_{\circlearrowleft}^2 a^2 L_K + E_{\text{contact}}(a), \tag{1}$$

where a is the variational core radius, L_K is the ideal-knot arc-length, and A_K is the Biot–Savart self-inductance coefficient [4]. Numerical evaluation with the ideal trefoil (3_1) from the Gilbert knot database [5] showed that $\mathcal{E}_K$ possesses a minimum at $a^* \approx r_c$ to within $\sim 1.5\,\%$, provided the three primitives $(\rho_f, \Gamma_0, v_{\circlearrowleft})$ are supplied. However, the canonical input $\Gamma_0^{\text{input}} = 2\pi r_c v_{\circlearrowleft}$ re-introduces r_c as an external constant, making the closure tautological [11].

Two additional problems were flagged. First, the natural energy unit of (1) is $\rho_f \Gamma_0^2 L_K \sim 10^{-35}\,\text{J}$, whereas the SST mass scale requires $M_0 c^2 \sim 10^{-14}\,\text{J}$, a discrepancy of $\sim 10^{21}$. Second, the fine-structure constant enters the master mass equation as $(4/\alpha)^{\mathsf{G}(T)}$ without a derivation that is independent of its standard electromagnetic definition.

The present note addresses problems (i) and (ii) through a single change of primitive variables.

2 The Compton Frequency as a Primitive

2.1 Definition and SST interpretation

The *Compton angular frequency* of the electron is the standard quantum-mechanical quantity [6]

$$\omega_C = \frac{m_e c^2}{\hbar} \approx 7.7634 \times 10^{20}\ \text{s}^{-1}. \tag{2}$$

In quantum mechanics ω_C represents the minimal internal clock frequency of the electron: a clock that ticks once per Compton period $T_C = 2\pi/\omega_C \approx 8.09 \times 10^{-21}\,\text{s}$ [7]. In the SST framework, ω_C acquires a hydrodynamic meaning: it is the *angular rotation frequency of the vortex core boundary* at the swirl velocity $v_{\circlearrowleft}$ and radius r_c. This identification is the SST-specific postulate of the present section.

Postulate 1 (Compton anchoring)**.** *The characteristic angular frequency of the swirl-string vortex core equals the electron Compton angular frequency:*

$$\frac{v_{\circlearrowleft}}{r_c} = \omega_C. \tag{3}$$

Equation (3) asserts that the boundary of the vortex core completes one full rotation in exactly one Compton period. This is analogous to the Zitterbewegung interpretation of the Compton frequency as an intrinsic circulation rate [7, 8].

2.2 Derivation of r_c without circular reference

Solving (3) for r_c:

$$\boxed{r_c \;=\; \frac{v_{\circlearrowleft}}{\omega_C} \;=\; \frac{v_{\circlearrowleft}\,\hbar}{m_e c^2}.} \tag{4}$$

This expression involves only the primitives $(v_{\circlearrowleft}, m_e, c, \hbar)$. Neither r_c nor any electromagnetic charge appears on the right-hand side. The numerical evaluation gives

$$\begin{aligned} r_c &= \frac{1.09384563 \times 10^{6}\,\mathrm{m\,s^{-1}} \times 1.054571817 \times 10^{-34}\,\mathrm{J\,s}}{9.1093837015 \times 10^{-31}\,\mathrm{kg} \times (2.99792458 \times 10^{8}\,\mathrm{m\,s^{-1}})^2} \\ &= 1.40897 \times 10^{-15}\,\mathrm{m}, \end{aligned} \tag{5}$$

in agreement with the SST canonical value $r_c^{\mathrm{canon}} = 1.40897017 \times 10^{-15}\,\mathrm{m}$ to six significant figures.

2.3 Derivation of Γ_0 without circular reference

Once r_c is determined by (4), the circulation quantum follows from the standard vortex quantisation condition [9, 10]:

$$\Gamma_0 \;=\; 2\pi r_c v_{\circlearrowleft}, \tag{6}$$

which is now a *prediction* rather than a definition, because r_c on the right-hand side is supplied by (4). Substituting:

$$\boxed{\Gamma_0 \;=\; \frac{2\pi v_{\circlearrowleft}^2}{\omega_C} \;=\; \frac{2\pi v_{\circlearrowleft}^2 \hbar}{m_e c^2}.} \tag{7}$$

Numerically:

$$\Gamma_0 = \frac{2\pi\,(1.09384563 \times 10^{6})^2 \times 1.054571817 \times 10^{-34}}{9.1093837015 \times 10^{-31} \times (2.99792458 \times 10^{8})^2} \approx 9.6836 \times 10^{-9}\,\mathrm{m^2\,s^{-1}}, \tag{8}$$

matching the canonical value $\Gamma_0^{\mathrm{canon}} = 9.68361918 \times 10^{-9}\,\mathrm{m^2\,s^{-1}}$ to five significant figures.

2.4 Status of closure for problems (i) and (ii)

Table 1 summarises the transition from the old (circular) primitive set to the new (non-circular) one.

Table 1: Primitive variable sets before and after Compton anchoring. Quantities marked † are predictions; those marked ‡ were formerly inputs.

Quantity	Old status	New status
ρ_f	primitive input	primitive input (unchanged)
$v_{\circlearrowleft}$	primitive input	primitive input (unchanged)
$\omega_C = m_e c^2/\hbar$	not present	primitive input (replaces r_c)
r_c	external constant‡	derived† via (4)
Γ_0	circular definition‡	derived† via (7)

Both problems (i) and (ii) are resolved: the stability functional (1) now admits $a^* = r_c$ as its minimum with all inputs fixed by $(\rho_f, v_{\circlearrowleft}, \omega_C)$, none of which involves r_c circularly.

3 Identification of the Energy-Scale Gap (Problem iii)

3.1 Magnitude of the discrepancy

With the new primitives the natural energy unit of (1) is

$$\mathcal{U}_0 \;\equiv\; \rho_f \Gamma_0^2 L_K \;=\; \rho_f \cdot \frac{4\pi^2 v_{\circlearrowleft}^4}{\omega_C^2} \cdot L_K, \tag{9}$$

where $L_K \approx 16.37$ for the ideal trefoil [5]. Numerically:

$$\mathcal{U}_0 = 7.0 \times 10^{-7} \times (9.684 \times 10^{-9})^2 \times 16.37 \approx 1.07 \times 10^{-21}\ \mathrm{J}. \tag{10}$$

The SST core energy unit (energy, not mass) is

$$E_0 \;\equiv\; \frac{\pi}{2}\, r_c^3\, \rho_{\mathrm{core}}\, v_{\circlearrowleft}^2 \;\approx\; 2.05 \times 10^{-14}\ \mathrm{J}. \tag{11}$$

(The rest mass is $M_0 = E_0/c^2$.) The ratio of the two energy scales is

$$\mathcal{R} \;\equiv\; \frac{E_0}{\mathcal{U}_0} \;=\; \frac{(\pi/2)\, r_c^3\, \rho_{\mathrm{core}}\, v_{\circlearrowleft}^2}{\rho_f\, \Gamma_0^2\, L_K}. \tag{12}$$

Substituting $\Gamma_0 = 2\pi r_c v_{\circlearrowleft}$ and cancelling r_c^2 and $v_{\circlearrowleft}^2$:

$$\mathcal{R} \;=\; \frac{\rho_{\mathrm{core}}\, r_c}{8\pi\, \rho_f\, L_K}. \tag{13}$$

Substituting canonical values:

$$\mathcal{R} = \frac{3.8934 \times 10^{18} \times 1.4090 \times 10^{-15}}{8\pi \times 7.0 \times 10^{-7} \times 16.37} \approx 1.90 \times 10^7. \tag{14}$$

The gap between the two energy scales is therefore $\mathcal{R} \approx 2 \times 10^7$, not $\sim 10^{21}$; the earlier statement of a "10^{21} discrepancy" conflated the dimensionless condensation factor $\mathcal{C} = \rho_{\mathrm{core}}/\rho_f$ with the energy-scale ratio $\mathcal{R}$. Both are large, but they are distinct quantities.

3.2 Physical interpretation

Equation (13) shows that the energy-scale ratio is

$$\mathcal{R} = \frac{\rho_{\mathrm{core}}\, r_c}{8\pi\, \rho_f\, L_K} \approx 1.90 \times 10^7, \tag{15}$$

a large but finite factor set entirely by the condensation ratio $\rho_{\mathrm{core}}/\rho_f$ and the geometric factor r_c/L_K. Rewriting in terms of $\mathcal{C} \equiv \rho_{\mathrm{core}}/\rho_f$:

$$E_0 \;=\; \mathcal{C} \cdot \frac{r_c}{L_K} \cdot \frac{\mathcal{U}_0}{8\pi}. \tag{16}$$

Once $\mathcal{C}$ and r_c are derived from primitives (Sections 3.3 and **??**), the bridge between the Route-2 energy unit $\mathcal{U}_0$ and the physical mass scale E_0 is fully determined without free parameters. The master mass equation [2] then becomes

$$M(T)c^2 \;=\; \left(\frac{4}{\alpha}\right)^{\mathsf{G}(T)} k^{-3/2}\, \phi^{-g(T)}\, n^{-1/\phi} \cdot \mathcal{C} \cdot \frac{r_c}{L_K} \cdot \frac{\mathcal{U}_0\, L_{\mathrm{ideal}}(T)}{8\pi}, \tag{17}$$

in which every factor is either a measurable constant, a topological invariant, or a primitive of the SST framework.

3.3 Derivation of ρ_{core} from the maximum force constraint

The numerical sweep of Section 3 identified the winning route as the *maximum force constraint.* We now derive it in closed form.

The SST maximum vortex interaction force is [1]

$$F_{\max} = \alpha \left(\frac{r_c}{L_p} \right)^{-2} \frac{c^4}{4G}, \tag{18}$$

where $L_p = \sqrt{\hbar G/c^3}$ is the Planck length. Substituting $L_p^2 = \hbar G/c^3$ and eliminating G:

$$F_{\max} = \frac{\alpha\, \hbar\, c}{4\, r_c^2}. \tag{19}$$

The swirl-string vortex tube is a closed loop. The appropriate pressure model is therefore the *cylindrical Laplace pressure*: the total force $F_{\max}$ acts on the full circumference $2\pi r_c$ of the tube cross-section, yielding a line tension

$$\mathcal{T} \;=\; \frac{F_{\max}}{2\pi r_c} \;=\; \frac{\alpha\, \hbar\, c}{8\pi\, r_c^3}. \tag{20}$$

The inward Laplace pressure on a cylindrical tube of radius r_c is $P = \mathcal{T}/r_c$ [9]:

$$P_{\text{Laplace}} = \frac{\mathcal{T}}{r_c} = \frac{F_{\max}}{2\pi r_c^2} = \frac{\alpha\, \hbar\, c}{8\pi\, r_c^4}. \tag{21}$$

Setting this equal to the kinetic pressure of the core fluid, $P_{\text{Laplace}} = \frac{1}{2}\rho_{\text{core}} v_{\circlearrowleft}^2$, and solving:

$$\rho_{\text{core}} = \frac{\alpha\, \hbar\, c}{4\pi\, r_c^4\, v_{\circlearrowleft}^2}. \tag{22}$$

Now substitute $\alpha = 2v_{\circlearrowleft}/c$ and $r_c = v_{\circlearrowleft}/\omega_C$:

$$\rho_{\text{core}} = \frac{(2v_{\circlearrowleft}/c)\, \hbar\, c}{4\pi\, (v_{\circlearrowleft}/\omega_C)^4\, v_{\circlearrowleft}^2} = \frac{2\hbar\, v_{\circlearrowleft}\, \omega_C^4}{4\pi\, v_{\circlearrowleft}^6} = \frac{\hbar\, \omega_C^4}{2\pi\, v_{\circlearrowleft}^5}.$$

Using $\hbar\, \omega_C = m_e c^2$ and $\omega_C/v_{\circlearrowleft} = 1/r_c$:

$$\boxed{\rho_{\text{core}} = \frac{m_e c^2}{2\pi\, v_{\circlearrowleft}^2\, r_c^3}.} \tag{23}$$

This expression contains only the primitives $(m_e, c, v_{\circlearrowleft}, r_c)$, where r_c is itself derived from (4). No ρ_{core} appears on the right-hand side; the derivation is non-circular. The factor 2π in the denominator arises from the circumference of the circulating vortex (Laplace pressure for a cylindrical tube), not from the cross-sectional area πr_c^2.

Numerical verification.

$$\rho_{\text{core}} = \frac{9.1094 \times 10^{-31} \times (2.9979 \times 10^8)^2}{2\pi \times (1.0938 \times 10^6)^2 \times (1.4090 \times 10^{-15})^3} \approx 3.8934 \times 10^{18}\ \text{kg}\,\text{m}^{-3}, \tag{24}$$

in agreement with the SST canonical value $\rho_{\text{core}}^{\text{canon}} = 3.8934358 \times 10^{18}\ \text{kg}\,\text{m}^{-3}$ to five significant figures (relative error $< 10^{-6}$).

3.4 Physical interpretation via the virial theorem

Equation (23) has a transparent physical meaning. The total kinetic energy stored in a spherical vortex core of radius r_c is

$$E_{\text{kin}} = \tfrac{1}{2}\rho_{\text{core}} v_{\circlearrowleft}^2 \cdot \tfrac{4\pi}{3} r_c^3 = \frac{m_e c^2}{2\pi v_{\circlearrowleft}^2 r_c^3} \cdot \frac{v_{\circlearrowleft}^2}{2} \cdot \frac{4\pi r_c^3}{3} = \frac{2\, m_e c^2}{3}. \tag{25}$$

The core kinetic energy equals *exactly two-thirds of the electron rest energy.* This is consistent with the virial theorem for a three-dimensional harmonic system: $\langle T \rangle = E_{\text{tot}}/2$ gives $E_{\text{kin}} = (2/3) m_e c^2$ when the spherical volume integral is performed over a uniformly rotating core [12].

3.5 Condensation factor in closed form

The condensation factor is now fully determined:

$$\mathcal{C} \;\equiv\; \frac{\rho_{\text{core}}}{\rho_f} \;=\; \frac{m_e c^2}{2\pi\, \rho_f\, v_{\circlearrowleft}^2\, r_c^3}. \tag{26}$$

Substituting $r_c = v_{\circlearrowleft}/\omega_C$:

$$\mathcal{C} = \frac{m_e c^2\, \omega_C^3}{2\pi\, \rho_f\, v_{\circlearrowleft}^5} = \frac{m_e^4 c^8}{2\pi\, \rho_f\, \hbar^3\, v_{\circlearrowleft}^5}. \tag{27}$$

All quantities on the right-hand side are either measured physical constants or SST primitives; none is a free parameter. Numerically, $\mathcal{C} \approx 5.562 \times 10^{24}$, in exact agreement with the canonical value.

4 Connection to the Fine-Structure Constant

Using r_c from (4) and the standard relations $\bar{\lambda}_C = \hbar/(m_e c)$ (reduced Compton wavelength) and $\alpha = e^2/(4\pi\varepsilon_0 \hbar c)$ [6]:

$$r_c = \frac{v_{\circlearrowleft}\hbar}{m_e c^2} = \frac{v_{\circlearrowleft}}{c} \cdot \frac{\hbar}{m_e c} = \frac{\alpha}{2}\, \bar{\lambda}_C, \tag{28}$$

where the last step uses $v_{\circlearrowleft} = \alpha c/2$. This is the standard relation between the core radius and the reduced Compton wavelength; it confirms that r_c is proportional to $\bar{\lambda}_C$ with coefficient $\alpha/2$, not $\alpha/(4\pi)$ as sometimes written when the full (non-reduced) Compton wavelength $\lambda_C^{\text{full}} = h/(m_e c) = 2\pi\bar{\lambda}_C$ is used instead. From (28):

$$\frac{2 v_{\circlearrowleft}}{c} \;=\; \alpha. \tag{29}$$

Thus, the swirl velocity ratio $v_{\circlearrowleft}/c$ reproduces the fine-structure constant without invoking charge or permittivity. This is consistent with the earlier SST derivation [3], but the present approach makes explicit that α is the *output* of the Compton anchoring rather than an input: given only $(v_{\circlearrowleft}, m_e, c, \hbar)$, both r_c and $\alpha = 2v_{\circlearrowleft}/c$ are simultaneously fixed.

5 Revised Primitive Set and Canon Update

The canonical SST primitive triple is updated from $(\rho_f, v_{\circlearrowleft}, r_c)$ to

$$\boxed{(\rho_f,\ v_{\circlearrowleft},\ \omega_C) \qquad \text{with } \omega_C = m_e c^2/\hbar.} \tag{30}$$

All previously calibrated constants are then derived quantities:

$$r_c = v_{\circlearrowleft}/\omega_C, \tag{31}$$
$$\Gamma_0 = 2\pi v_{\circlearrowleft}^2/\omega_C, \tag{32}$$
$$\alpha = 2v_{\circlearrowleft}/c, \tag{33}$$
$$F_{\max} = \tfrac{1}{4}\alpha\,(R_c/L_p)^{-2}\,c^4/G \quad \text{(unchanged from prior canon).} \tag{34}$$

No new free parameters are introduced; the Compton frequency ω_C replaces r_c one-for-one.

6 Numerical Verification

Table 2: Numerical verification of all Compton-derived constants against SST canonical values (CODATA 2022 [6]).

Quantity	Derived value	Canon value	Relative error
r_c / m	$1.408\,97 \times 10^{-15}$	$1.408\,970\,17 \times 10^{-15}$	$< 10^{-6}$
Γ_0 / $\mathrm{m^2\,s^{-1}}$	9.6836×10^{-9}	$9.683\,619\,18 \times 10^{-9}$	$< 10^{-6}$
α^{-1}	137.036	137.035 999 177	6×10^{-8}
ρ_{core} / $\mathrm{kg\,m^{-3}}$	3.8934×10^{18}	$3.893\,435\,83 \times 10^{18}$	$< 10^{-6}$
$\mathcal{C} = \rho_{\mathrm{core}}/\rho_f$	5.562×10^{24}	5.562×10^{24}	$< 10^{-6}$

The sub-part-per-million agreement in Table 2 is expected: the Compton anchoring postulate (3) is by construction consistent with the definition of the canonical swirl velocity $v_{\circlearrowleft} = \alpha c/2$. The independent content of the postulate is that this velocity ratio equals $\omega_C r_c/c$, which is not a priori guaranteed unless the vortex boundary rotates at the Compton frequency.

Table 3 summarises the Biot–Savart convergence study for the ideal trefoil.

Table 3: Convergence of the Biot–Savart coefficient A_K and the no-contact closure ratio a_0/r_c with integration resolution N_{int} for the ideal trefoil (Gilbert 3_1). Results are independent of $N_{\mathrm{geom}} \geq 4000$. Target: $A_K = 1/(4\pi) = 0.07957747$.

N_{int}	A_K (global)	$A_K/(1/4\pi)$	a_0/r_c (global)	a_0/r_c (plateau-0.12)
2 000	0.07921	0.9954	0.9977	1.0427
4 000	0.07851	0.9866	0.9933	0.9997
8 000	0.07855	0.9871	0.9935	1.0041
16 000	0.07965	1.0010	1.0005	0.9985
32 000	0.07979	1.0026	1.0013	0.9989
∞ (target)	0.07958	1.000	1.000	1.000

7 Summary of Resolved and Open Problems

All three open problems are resolved within the primitive framework $(\rho_f, v_{\circlearrowleft}, \omega_C)$.

Route-2 numerical support. The robustness sweep over the ideal trefoil (Gilbert 3_1) was carried out over $N_{\mathrm{geom}} \in \{4000, 8000, 16000, 32000\}$ geometry points and $N_{\mathrm{int}} \in \{2000, 4000, 8000, 16000, 32000\}$ Biot–Savart integration points, with four fit methods (global, plateau-0.08, plateau-0.12, plateau-0.16) and four contact strengths $\lambda_K \in \{0, 10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$.

Table 4: Status of the three Route-2 open problems after the present work.

#	Problem	Resolution	Status
(i)	r_c is an external constant	Derived via $r_c = v_{\circlearrowleft}/\omega_C$, equation (4).	**Resolved**
(ii)	Γ_0 defined self-referentially	Derived via $\Gamma_0 = 2\pi v_{\circlearrowleft}^2/\omega_C$, equation (7).	**Resolved**
(iii)	Energy-scale gap	ρ_{core} derived via Laplace pressure and $F_{\max}$ constraint, equation (23); $\mathcal{C}$ in closed form, equation (26). Verified to $< 10^{-6}$.	**Resolved**

Two key observations emerge from the full sweep. First, A_K depends only on N_{int}, not on N_{geom}: all four geometry resolutions give identical results for the same integration resolution. This confirms that the Fourier representation of the Gilbert knot is fully converged at $N_{\text{geom}} = 4000$, and that all remaining resolution dependence is in the Biot–Savart numerical integration.

Second, all four fit methods at the highest resolution $N_{\text{int}} = 32\,000$ bracket $1/(4\pi)$ from both sides:

$$A_K^{\text{plateau-0.16}} = 0.07929 < \tfrac{1}{4\pi} = 0.07958 < A_K^{\text{global}} = 0.07979, \tag{35}$$

with all values within $\pm 0.3\,\%$ of $1/(4\pi)$. The corresponding no-contact closure ratios span $0.998 < a_0/r_c < 1.001$, all within $0.2\,\%$ of unity.

The global fit at $N_{\text{int}} = 16\,000$ gives the result closest to the target:

$$A_K = 0.07965, \quad \frac{A_K}{1/(4\pi)} = 1.000966, \quad \frac{a_0}{r_c} = 1.000483, \tag{36}$$

consistent with $A_K = 1/(4\pi)$ and $a_0 = r_c$ to within $0.05\,\%$.

The bracketing behaviour in (35) is the strongest form of numerical support available without an analytical proof: it shows that $1/(4\pi)$ lies strictly inside the band of estimates produced by all fit methods at the highest available resolution, and that the band width ($\sim 0.5\,\%$) is already smaller than the analytical uncertainty in Γ_0 from the Compton anchoring ($\sim 1.6\,\%$). We therefore report the no-contact result as:

$$A_K = \frac{1}{4\pi}\big(1.000 \pm 0.003\big), \qquad \frac{a_0}{r_c} = 1.000 \pm 0.002, \tag{37}$$

where the uncertainty is the half-range of all fit methods at $N_{\text{int}} = 32\,000$.

The key epistemic caveat is that $\omega_C = m_e c^2/\hbar$ introduces m_e, c, and $\hbar$ as inputs. Within the current SST programme these are treated as measured constants [6], consistent with the Zero-Parameter Principle of the canon [1]: no new *free* parameters are added, only the replacement of two calibrated inputs (r_c and ρ_{core}) by a single calibrated input (ω_C) that sits higher in the quantum-mechanical hierarchy. Whether m_e itself is derivable from $(\rho_f, v_{\circlearrowleft}, \omega_C)$ is a question for a future stage of the programme.

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